

KuantorFlow — Weekly Development Report

Period: 18–24 July 2026 · Repositories: kuantorflow, ai_agent, kuantorflow_automation

Executive Summary

This week finished and hardened the **Settings platform**, grew **Mykola** into a more capable companion, and added genuinely new **study and content** features — a flip-card deck, a Reverso copy-paste parser with AI-assisted translation splitting, and an early duplicate-word warning. **19 pull requests merged** across the three repositories, and the regression suite grew to **114 passing offline tests** — every feature now shipped with a parallel test PR and, for the significant ones, a committed test report.

Repository	Role this week	Merged PRs
kuantorflow	Settings hardening, study/content features, UI polish	11
kuantorflow_automation	Test PRs, the test-report convention, suite → 114	7
ai_agent	Mykola identity	1
Total		19

Per the standing request, commit authorship was checked across all three repositories for the period: every commit was authored by Kuantor (Claude as co-author). No GitHub Copilot commits were found in this range.

Completed Work by Theme

1. Settings platform — finished and hardened

The remaining Highest-Prio settings items all landed:

- **Reset Auth** (#98, PR #117) — a button that clears the whole session (keyword gate + Google identity, both held in the signed cookie) and the app's browser storage, returning to the gate. Enabled even inside the read-only popup, since it's an action, not a setting.
- **Anonymous settings frozen** (#102, PR #116) — anonymous visitors share config-default.json, so one of them must not change it for everyone: POST /settings now refuses them (403) and the popup renders read-only with a sign-in prompt.
- **Two-column popup layout** (#118, PR #119) — the popup had grown past the viewport; the fieldsets now sit in a responsive two-column grid, the actions share one row, and the dialog clamps to the screen.
- **Mobile widget hidden with the popup** (#121, PR #122) — on phones the Mykola widget no longer overlaps the open Settings popup.

2. Mykola the companion

- **"New Chat" button** (ai_agent#55, PR #128) — a small pencil in the widget header starts a fresh conversation behind a Yes/No confirmation, re-running the welcome-back recap for signed-in users; past exchanges stay logged.
- **Claude-powered identity** (ai_agent#48, PR #53) — Mykola now answers "Are you Claude?" honestly and in character, while still giving his playful symbolic age.

3. Richer study & content

- **Card deck activity** (#78, PR #135) — a Quizlet-style flip-card deck per topic: one card at a time, click/Enter/Space to flip, arrow keys to move.
- **Reverso copy-paste parser** (#134, PR #141) — .mht notes that are OneNote copy-pastes of Reverso entries are auto-detected and parsed into rich cards (one per word + part of speech, senses aggregated, with examples). Reverso glues the translation terms together with no separators, so **Claude splits them back apart**, keeping multi-word phrases intact, with a graceful fallback when no API key is present.
- **Early duplicate-word warning** (#145, PR #149) — looking up a word that already has cards now warns *before* the lookup and review dialog ("look it up anyway?"), complementing the save-time duplicate skip from #101.

4. UI polish & docs

- **Blue links panel** (#142, PR #143) — the "Home | Card deck | Take quiz" menu on the flashcards, deck and quiz pages is now a blue rounded panel.
- **Live website link in the README** (#115, PR #123; PR #124) — with access instructions.

5. Process & quality (kuantorflow_automation)

- **Test-report convention** (#103, PR #18) — significant PRs now get a Markdown + PDF verification report committed under test_reports/, rendered with the report tooling; small PRs are exempt unless requested.
 - **Seven test PRs** covered every feature above; the offline suite grew from ~100 to **114 passed**, still under three seconds and fully offline.
 - The **parallel feature-and-test PR** practice (#15) was confirmed and recorded.
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Technical Highlights

- **Format detection as a state machine.** The Reverso parser recognises the copy-paste format by its colour-coded markup and walks the paragraphs as a header → sense → explanation/example/translation state machine — robust to the fixed OneNote export shape.
- **AI where the markup can't help.** Reverso concatenates translation terms with no separators, so they cannot be split from the HTML; a single batched Claude (haiku) call splits them, keeping phrases like "верховный правитель" whole — and degrading to the whole string when offline.
- **Graceful degradation throughout.** The translation split, and the duplicate-word check, both treat a missing API key or an unreachable DB as "unknown" and proceed, so no feature hard-fails in a degraded environment.

- **One escaping bug worth noting.** Threading parsed examples through the review popup exposed that Jinja's tojson output isn't attribute-safe on its own (its quotes break value="..."); tojson | forceescape fixes it.
- **Session-only auth model paid off.** Because the gate pass and Google identity both live in the signed session cookie, Reset Auth is a one-line session.clear() — no tokens to track.

Lessons Learned

1. **`tojson` needs `forceescape` inside an HTML attribute.** Its output is HTML-safe for text nodes but its literal " characters break a double-quoted attribute value.
2. **Design for the degraded path first.** Both the AI translation split and the duplicate warning were built to work with no API key / no database, which also made them trivial to test offline.
3. **Confirm destructive-looking actions, but keep them cheap.** New Chat and Reset Auth both got Yes/No dialogs; neither is actually destructive (chats stay logged, settings files survive), which is worth stating in the copy so users aren't scared off.
4. **Board access needs the right token.** Reading the "KuantorFlow Improvements" board needs the GitHub project scope; it lives on the gh CLI's own keyring login, so the board is queried with gh's auth rather than the repo PAT used for pushes. With that in place, the plan below reflects the real board columns.

Plans for Next Week

Taken from the KuantorFlow Improvements GitHub Projects board ([dashboard] (<https://github.com/users/Kuantor/projects/2>)), in board-column order: Highest Prio and Nice To Have first, then In Progress and Todo.

Highest Prio

Repo	Issue	Title
kuantorflow	#30	Create logs for various actions in the app
kuantorflow	#45	UI: enhance behaviour of MHT-processing results window
kuantorflow	#100	Quiz: treat perfective/imperfective answers as equal (AI)
kuantorflow	#137	Upload notes: support .txt and .docx besides .mht
kuantorflow	#138	Develop engaging flashcard topics for the online database
kuantorflow	#144	Add Ukrainian/Russian translations after parsing MHT files
kuantorflow	#146	Common "Cancel" cross for closing the new-card popup
kuantorflow	#147	More accurate classification of words in the "General" topic + AI
ai_agent	#19	Teach Mykola to complement/extend card meanings
ai_agent	#54	Restart Mykola's conversations frequently

Nice To Have

Repo	Issue	Title
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kuantorflow	#22	Redesign website for a modern look
kuantorflow	#92 / #94	Spaced-repetition review mode (SM-2)
kuantorflow	#93	Complete multi-user card ownership
kuantorflow	#125	Deny unauthorized users to add/change/delete words
kuantorflow	#126	Blacklist file for blocking users
kuantorflow	#127	Create "Individual" cards mode
kuantorflow	#129	Smart word-manipulation techniques for exercises (spike)
kuantorflow	#136	Images on flashcards (files on disk, path in MySQL)
kuantorflow	#139	Delete-word request to the admin
kuantorflow	#85	Full-screen widget mode with visible maximize button
kuantorflow	#88 / #89 / #90	Backup docs / added-by field / logs disclosure
kuantorflow	#99 / #104	API-credits message / Allura welcome font
ai_agent	#51 / #56 / #57 / #58	Nicknames / bigger avatar / add-card confirm / save conversation
kuantorflow_automation	#4	Basic automated tests for the Mykola AI agent

In Progress — a new "word games" cluster

Repo	Issue	Title
kuantorflow	#130	Levenshtein multiple-choice translation game
kuantorflow	#131	Keyboard-adjacency typo distractors (MCQ)
kuantorflow	#132	"Is this a real word?" (n-gram pseudowords)
kuantorflow	#133	Typoglycemia scrambling mode
kuantorflow	#110	Investigation: Learner's vs Collegiate dictionary (+ M-W API)

Todo (selected)

Repo	Issue	Title
kuantorflow	#74	Dark theme for the website
kuantorflow	#25	Export/import cards (local ↔ remote DB)
kuantorflow	#148	IDs for users
kuantorflow	#120	Pin the Settings close button while the dialog scrolls
kuantorflow	#140	[Optional] Upload notes: support .pdf (text-layer)
kuantorflow	#19 / #75 / #76 / #84	Tutorial tip / About-popup overlap / scheduled recaps / dim welcome
ai_agent	#47 / #49 / #50	Widget-code audit / last-logs setting / faster Mykola

Report generated 24 July 2026 from GitHub PR, issue and commit data across the three repositories. Merged-PR window: 18 July 2026 (kuantorflow#116/#117) through 24 July 2026 (kuantorflow#149, kuantorflow_automation#21).